10.4 Inscribed Angles and Polygons

Name _____ Period ____ Date _____

Geometry 10.4 (30-minute plan) · Student sheet

I can…

- find the measure of an inscribed angle or its intercepted arc.
- use the fact that a triangle inscribed in a semicircle is a right triangle.
- use the fact that opposite angles of an inscribed quadrilateral are supplementary.

Vocabulary · fill in the missing word

Inscribed angle — an angle whose vertex is on a circle and whose sides _____ chords of the circle

Intercepted arc — the arc that lies in the interior of an _____ angle and has endpoints on the angle

Inscribed polygon — a polygon whose _____ all lie on a circle

Circumscribed circle — the circle that contains all the vertices of an _____ polygon

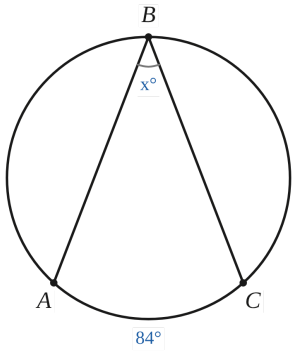
Part 1 · Notes and examples (fill in each note, then work the example)

Measure of an Inscribed Angle Theorem: the measure of an inscribed angle is _____ the measure of its intercepted arc.

Example 1

a. Find m∠ABC.

b. An inscribed angle measures 55°. What is the measure of its intercepted arc?

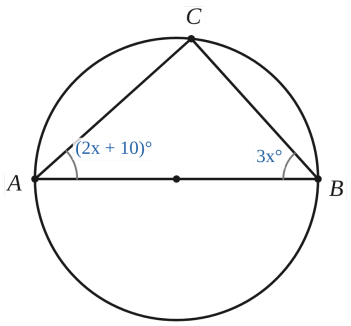


Inscribed Angles of a Circle Theorem: if two inscribed angles of a circle intercept the same arc, then the angles are _____.

Inscribed Right Triangle Theorem: a triangle inscribed in a circle is a right triangle if and only if one side is a _____.

Example 2

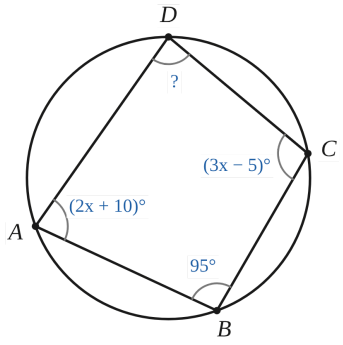
$\overline{AB}$ is a diameter of the circle. Find the value of x. Then find m∠A and m∠B.



Inscribed Quadrilateral Theorem: a quadrilateral can be inscribed in a circle if and only if its opposite angles are _____.

Example 3

Quadrilateral ABCD is inscribed in the circle. Find the value of x. Then find m∠D.



Part 2 · Partner practice

Partner practice

Quadrilateral PQRS is inscribed in a circle. m∠Q = (4x + 8)° and m∠S = (2x + 10)°. Find x, m∠Q, and m∠S.